

Medhansh Rath

M. Sc. ROBOTICS STUDENT

As a highly motivated and academically successful M. Sc. Robotics, Cognition and Intelligence student, I have gained valuable practical experience in Computer Vision, Cobots, and Mobile Robotics. Constantly striving to expand my knowledge and skills, I am currently working on rover navigation. I enjoy applying what I have learned in the real world and have been awarded for my outstanding performance in several competitions, most recently the Best Transferability Prize in the euROBIN MSV Challenge at IROS 2024.

DETAILS



medhansh-rath
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SKILLS

Frameworks

ROS 1 & 2 Pytorch Keras

Coding

Python C/C++ Java
MATLAB Simulink

Software

Linux Git Docker

Prototyping

CAD 3D Printing

Hardware

Electronics Microcontrollers

LANGUAGES

English Native Level
German Fluent

ACHIEVEMENTS

Best Transferability Demo in
euROBIN MSVC, IROS 2024^{news}

Winner of Munich_i Robothon®
2023 by TUM & Automatica^{news}

Best Circular Impact at the
robo.innovate Circular Hackfest

Winner of Disrupt2Win 2023^{news}

National Rank 5 in ThinkLABS
Robotics Competition

SOF's IMO International Rank 10

EXPERIENCE

Working Student - Rovers

Oberpfaffenhofen, Germany

 GERMAN AEROSPACE CENTER

05.2026 – today

- Creating a IsaacSim environment for prototype rovers
- Testing global planning for the Idefix rover

Research Assistant - Computer Vision

Schweinfurt, Germany

 thws PROF. DR. VOLKER WILLERT

12.2025 – 09.2025

- Designed a novel superpixel algorithm for segmenting RGBD images as a preprocessing step for Vision Transformers as a continuation of my thesis

Working Student - Cobots

Munich, Germany

SIEMENS CORPORATE RESEARCH

02.2024 – 09.2024

- Prepared a high-visibility GPT-assisted robotic pick-and-place demo at the Siemens Technology Center inauguration^{news}
- Programmed multi-robot manipulation with UR 5 and 10 robots
- Extended Digital Twin models in a research environment

Intern - Computer Vision and NLP

Munich, Germany

SIEMENS CORPORATE RESEARCH

09.2023 – 01.2024

- Independently executed the analysis, design, and implementation of Product Manufacturing Information (PMI) extraction from schematics
- Automated the creation of labelled datasets, and finetuned Computer Vision models (YOLO, Segment Anything, Tesseract OCR)

- Implemented Retrieval Augmented Generation as part of the Siemens Industrial Copilot using LlamaIndex and GPT

- Developed a training pipeline for Large Language Models (LLama 2, Falcon) using Human Feedback Reinforced Learning (RLHF)

Scientific Assistant - Computer Vision + Tutor

Schweinfurt, Germany

 thws CENTER OF ROBOTICS AND INTELLIGENCE

10.2022 – 07.2023

- Trained models for GPU and RAM chip detection for E-Waste recycling
- Implemented detection of palletized cartons in an industrial scenario handling suboptimal lighting conditions and varying 3D perspectives

- Tutor for Mathematics 1 & 2 and Electronics & Components

- Bilingually assisted students as a part of the International Student Office

Intern - Test Automation

Munich, Germany

 Telefónica

07.2019

- Automated testing of CRM software with Tosca in an Agile environment
- Telefónica Certificate of Excellence for outstanding accomplishments

SCHOLARSHIPS

Studienstiftung des deutschen Volkes - Max Weber Programm

e-fellows.net Scholarship

PATENTS

[Application in EU, US, India]
Disassembly Assistance:
Methods And Systems^{link}

CERTIFICATES

German C1^{link}

Bilingual Bachelor Degree

TestAS Aptitude Exam (Top 1%)

Harvard CS50 Certificate

MITx Certificate for Mechanics^{link}

UK Mathematics Trust (Gold
Certificate & Certificate for Merit)

REFERENCES

Prof. Dr.-Ing. Volker Willert,
Center of Robotics and
Intelligence, THWS
volker.willert@thws.de

Prof. Dr. Kai Diethelm,
Head of Laboratory for
Scientific Computing, THWS
kai.diethelm@thws.de

EDUCATION

M. Sc. Robotics, Cognition, Intelligence

Munich, Germany

 TECHNICAL UNIVERSITY OF MUNICH

10.2025 – today

- Invited to sponsored hackathon by Sebastian Thrun for a stealth startup
- Subjects: Autonomous Vehicles, Cobots

B. Eng. Robotics (Grade: 1.3 | 1.0 Best)

Schweinfurt, Germany

 TH WÜRZBURG-SCHWEINFURT

10.2021 – 06.2025

- Salutatorian Award^{news}
- Thesis: Designing Perspective Visual Words for Vision Transformers
- Specialization: Mobile Robotics - SLAM, Drones, Mobile Platforms

IB Diploma (Grade: 1.1 | 1.0 Best)

Munich, Germany

 BAVARIAN INTERNATIONAL SCHOOL

08.2019 – 07.2021

- Higher Level: Math, Physics, Chemistry, German | Standard: History, English

PROJECTS

Rock-E

 github.com/medhansh-rath/rock_e

MARS ROVER FOR 3D RECONSTRUCTION

04.2024 – 02.2025

- Created a prototype Unmanned Ground Vehicle (UGV)
- Implemented autonomous navigation to outcrops using Visual SLAM
- Generated RGB point clouds using a LiDAR and a RGBD camera

euROBIN MSV Challenge

 github.com/medhansh-rath/MSVC2024

IROS 2024, ABU DHABI

09.2024 – 10.2024

- Refined and generalized taskboard detection
- Implemented solutions to tasks relevant to disassembly of electronics
- Set up a low-cost and eco-friendly demo of smartphone disassembly using eco 3D printed parts, suction pads, and a discarded phone and hairdryer
- Performed robot and camera calibration on an UR10e + Intel Realsense

CircularAR

[Link to Patent Application](#)

AI AND AR ASSISTED LAPTOP DISASSEMBLY

11.2023

- Prototyped an app combining GPT4-Vision with AR for laptop disassembly
- Pitched the idea at the Circular Hackfest '23 and won the popular vote prize

Lifeline Medical

[Kickstarter Idea](#)

FALL ARREST FOR HOSPITALS

02.2023 – 06.2023

- Designed a smart robotic harness for the safety of dementia patients in hospital environments and pitched in a kickstarter competition

OPEN SOURCE CONTRIBUTIONS

pointcloud

 github.com/medhansh-rath/pointcloud

POINTCLOUD CONVERSION WITH CUDA

New library

DASP

 github.com/Danvil/dasp

DEPTH ADAPTIVE SUPERPIXELS

Fixed old dependencies

CustomTkinter

 github.com/TomSchimansky/CustomTkinter

PYTHON UI LIBRARY

Added a new UI element class

HP OMEN RGB CLI

 github.com/OmenLinux/omen-rgb-keyboard-cli

RGB KEYBOARD CLI

Created installation script